Measles Susceptibility in Glasgow Primary School Children

DAS Group 15

2026-03-16

1 Introduction

The measles, mumps and rubella (MMR) vaccine has been an important part of childhood immunisation programmes in the UK since the late 1980s. In 1998, researcher Andrew Wakefield published a paper claiming a link between the MMR vaccine and autism. Although the study was later found to be fraudulent and was fully retracted in 2010, the media coverage it received caused vaccination rates to drop significantly, falling to around 80% coverage in some parts of the UK by 2003. This eventually contributed to a large measles outbreak in 2013.

A partial retraction of the paper occurred in 2004, which helped public perception of the vaccine. This report uses data from the Scottish Childhood Immunisation Record System (SCIRS) to investigate whether measles susceptibility among primary school children in Glasgow changed over the period 1998–2012, and whether any change can be linked to the 2004 partial retraction of the paper.

The data cover 133 intermediate zones (IZs) in Glasgow, which are small geographic units of roughly 4,000 residents each. For each IZ and year, we have the number of primary school children susceptible to measles (`Y`) and the total number of primary school children (`N`).

The two questions we aim to answer are:

1. Did measles susceptibility in Glasgow change following the Wakefield retraction?
2. Did this change occur specifically around 2004?

2 Exploratory Data Analysis

Table 1: Summary statistics for the proportion of primary school children susceptible to measles, by year.

		Proportion susceptible			
Year	IZs	Mean	SD	Min	Max
1998	133	0.0488	0.0322	0.0104	0.2581
2000	133	0.0599	0.0466	0.0000	0.3676
2002	133	0.0725	0.0435	0.0137	0.2734
2004	133	0.0680	0.0331	0.0000	0.2203
2008	133	0.0811	0.0504	0.0000	0.2787
2010	133	0.0518	0.0334	0.0000	0.1714
2012	133	0.0474	0.0328	0.0000	0.1923

Table 2: Summary statistics for the proportion of primary school children susceptible to measles, by year.

	Percenatge susceptible (%)			
Year	Mean	SD	Min	Max
1998	4.88	3.22	1.04	25.81
2000	5.99	4.66	0.00	36.76
2002	7.25	4.35	1.37	27.34
2004	6.80	3.31	0.00	22.03
2008	8.11	5.04	0.00	27.87
2010	5.18	3.34	0.00	17.14
2012	4.74	3.28	0.00	19.23

Table 1 shows how the mean proportion of susceptible children changed over time. There is an increase from 1998 through to around 2003, followed by a decline from 2004 onwards, which lines up with the timing of the Wakefield retraction.

Figure 1 shows the overall pattern of susceptibility rising from 1998, peaks around 2003, then falls after 2004. This is consistent with the drop in vaccination rates following the 1998 paper, and a recovery in uptake after the retraction. However, the change in the overall trend does not occur in 2004 directly. There appears to be an overall increase in susceptibility until 2008. That is, there appears to be a delayed effect.

Figure 2 shows the distribution across IZs for each year. There is moderate variation between IZs, which makes sense given that different areas will have different vaccination uptake.

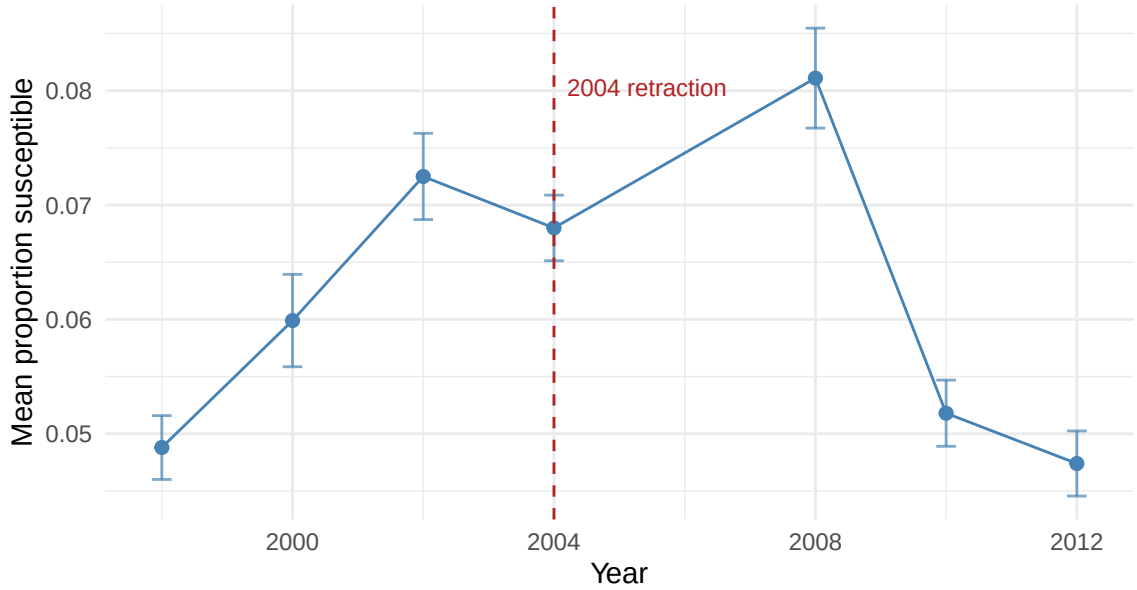


Figure 1: Mean proportion susceptible per year across all 133 Glasgow IZs. Error bars show +/- 1 standard error. The dashed line marks 2004.

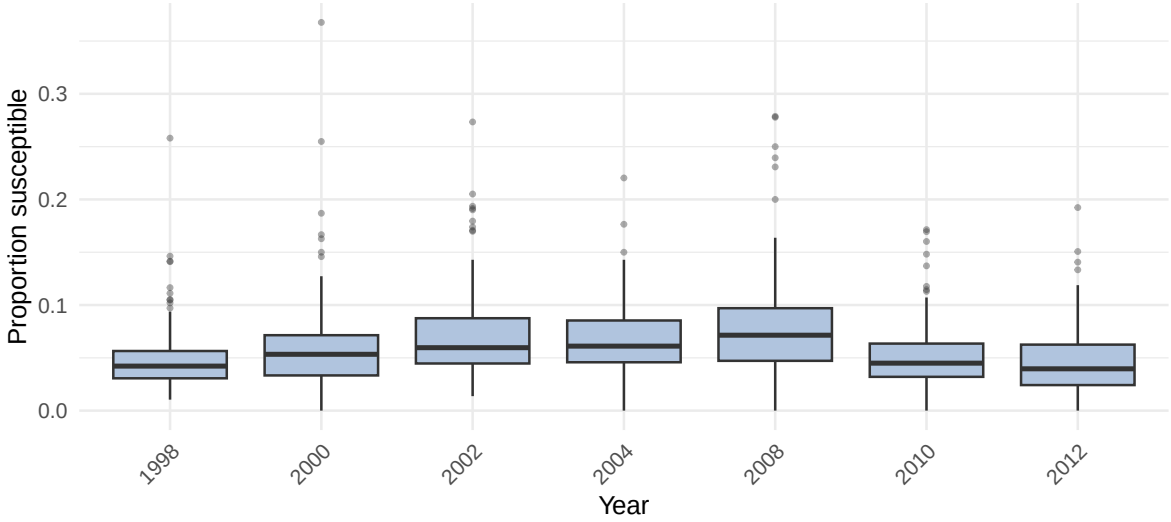


Figure 2: IZ-level proportion susceptible by year. Each box summarises the 133 IZs for that year.

3 Formal Analysis

3.1 Model

Since our outcome is a count of susceptible children out of a total (i.e., Y out of N), the model proposed is a binomial GLM with a logit link, which is a logistic regression model but for proportional count data rather than individual 0/1 outcomes.

We want to test whether the trend in susceptibility changed at 2004, so we fit a model with an interaction term between year and the post-2004 indicator:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \cdot \text{Year}_i^* + \beta_2 \cdot \mathbb{1}[\text{post}] + \beta_3 \cdot (\text{Year}_i^* \times \mathbb{1}[\text{post}])$$

where Year^* is the year centred at 2004, and $\mathbb{1}[\text{post}]$ equals 1 for 2004 and later. The interaction β_3 captures whether the slope differs before vs after 2004. Here, π_i is the probability of a child being susceptible to measles in year level i.

Fitting the this proposed model produced the following summary output.

Table 3: Table 2: Model summary for binomial model centred at 2004.

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.10	**<0.001**
yr c	1.12	**<0.001**
post	0.81	**0.001**
yr c × post	0.85	**<0.001**
Observations	931	
Deviance	2044.480	
AIC	4562.514	

```
Dispersion ratio: 2.205


# Overdispersion test

 dispersion ratio =   1.849
          p-value = < 0.001
```

The residual deviance of the binomial model (2044.5 on 927 df) gives a dispersion ratio of approximately 2.2, which exceeds the expected value of 1 under a correctly specified binomial

model. This suggests the presence of overdispersion, meaning the standard errors from the binomial model may be too narrow.

Table 4: Table 3: Model summary for binomial model centred at 2008.

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.10	**<0.001**
yr c	1.06	**<0.001**
post	0.81	**0.002**
yr c × post	0.81	**<0.001**
Observations	931	
Deviance	2002.928	
AIC	4529.192	

Table 5: Table 4: Odds Ratios for Year-on-Year Susceptibility Relative to 1998 Baseline

	cbind(Y,N-Y)	
Predictors	Odds Ratios	p
(Intercept)	0.05	**<0.001**
Year [2000]	1.26	**<0.001**
Year [2002]	1.57	**<0.001**
Year [2004]	1.42	**<0.001**
Year [2008]	1.70	**<0.001**
Year [2010]	1.09	0.118
Year [2012]	0.98	0.694
Observations	931	
Deviance	1966.583	
AIC	4508.119	

```
Call:
glm(formula = cbind(Y, N - Y) ~ factor(Year), family = binomial,
    data = dat)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      -3.01280    0.04157 -72.469  < 2e-16 ***
factor(Year)2000  0.22779    0.05607   4.063 4.85e-05 ***
factor(Year)2002  0.44954    0.05391   8.338  < 2e-16 ***
factor(Year)2004  0.35326    0.05531   6.387 1.69e-10 ***
```

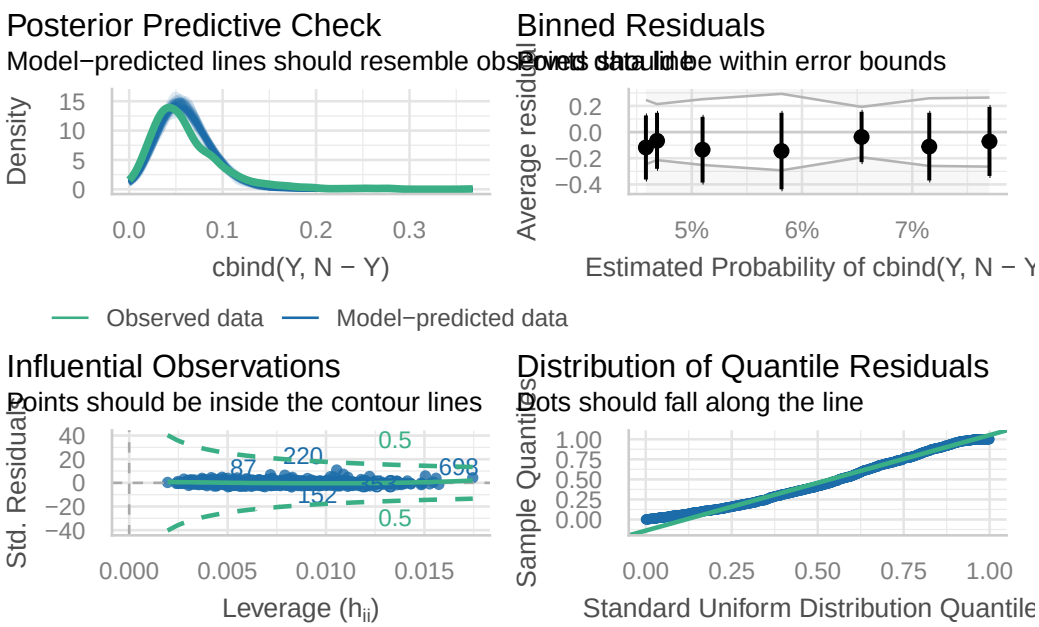
```
factor(Year)2008  0.52898    0.05306   9.970  < 2e-16 ***
factor(Year)2010  0.08885    0.05690   1.562    0.118
factor(Year)2012 -0.02281    0.05793  -0.394    0.694
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2179.6  on 930  degrees of freedom
Residual deviance: 1966.6  on 924  degrees of freedom
AIC: 5084.6

Number of Fisher Scoring iterations: 4


[1] 3.231932e-43
```



4 Consclusion